

[Vitamin A and Carotenoids: Fact Sheet for Consumers \(original version\)](#)

Vitamin A and Carotenoids: Fact Sheet for Consumers

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What are vitamin A and carotenoids and what do they do?

Vitamin A is a fat-soluble vitamin that is naturally present in many foods. Vitamin A is important for normal vision, the immune system, reproduction, and growth and development. Vitamin A also helps your heart, lungs, and other organs work properly. Carotenoids are pigments that give yellow, orange, and red fruits and vegetables their color. Your body is able to convert some carotenoids into vitamin A.

There are two different sources for vitamin A:

- **Preformed vitamin A** is found in fish, organ meats (such as liver), dairy products, and eggs.
- **Provitamin A carotenoids** are turned into vitamin A by your body. They are found in fruits, vegetables, and other plant-based products. The most common provitamin A carotenoid in foods and dietary supplements is beta-carotene.

How much vitamin A do I need?

The amount of vitamin A you need depends on your age and sex. Average daily recommended amounts of preformed vitamin A and provitamin A carotenoids are listed below in micrograms (mcg) of retinol activity equivalents (RAE).

Life Stage	Recommended Amount
Birth to 6 months	400 mcg RAE
Infants 7–12 months	500 mcg RAE
Children 1–3 years	300 mcg RAE
Children 4–8 years	400 mcg RAE
Children 9–13 years	600 mcg RAE
Teen males 14–18 years	900 mcg RAE
Teen females 14–18 years	700 mcg RAE
Adult males	900 mcg RAE

Life Stage	Recommended Amount
Adult females	700 mcg RAE
Pregnant teens	750 mcg RAE
Pregnant women	770 mcg RAE
Breastfeeding teens	1,200 mcg RAE
Breastfeeding women	1,300 mcg RAE

What foods provide vitamin A?

Vitamin A is found naturally in many foods and is added to some foods, such as milk and cereal. You can get recommended amounts of vitamin A by eating a variety of foods, including the following:

- Some types of fish, such as herring and salmon
- Beef liver and other organ meats (which are also high in cholesterol, so limit the amount you eat)
- Green leafy vegetables and other green, orange, and yellow vegetables such as spinach, sweet potatoes, carrots, broccoli, and winter squash
- Fruits, including cantaloupe, mangos, and apricots
- Dairy products, such as milk and cheese
- Fortified breakfast cereals
- Eggs

What kinds of vitamin A dietary supplements are available?

Vitamin A is available in dietary supplements, usually in the form of retinyl acetate or retinyl palmitate (preformed vitamin A), beta-carotene (provitamin A), or a combination of preformed and provitamin A. Most multivitamin/mineral supplements contain vitamin A. Dietary supplements that contain only vitamin A are also available.

Am I getting enough vitamin A?

Vitamin A deficiency is rare in the United States because most people get enough vitamin A from the foods they eat. However, vitamin A deficiency is common in many developing countries, especially in young children.

Certain groups of people are more likely than others to have trouble getting enough vitamin A:

- Premature infants
- Infants, young children, pregnant women, and breastfeeding women in developing countries
- People with cystic fibrosis
- People with Crohn's disease, ulcerative colitis, or celiac disease

What happens if I don't get enough vitamin A?

The most common sign of vitamin A deficiency is an eye condition called xerophthalmia. Xerophthalmia is the inability to see in low light, and it can lead to blindness if it isn't treated.

A long-term deficiency of vitamin A can also lead to a higher risk of respiratory diseases (such as pneumonia) and infections (such as measles and diarrhea). It can also cause anemia (a condition in which the red blood cells do not supply enough oxygen to the body). In severe cases, not getting enough vitamin A can increase your chances of dying.

What are some effects of vitamin A on health?

Scientists are studying vitamin A to understand how it affects health. Here are some examples of what this research has shown.

Cancer

People who eat a lot of foods containing vitamin A or beta-carotene might have a lower risk of certain kinds of cancer. However, studies do not show that vitamin A or beta-carotene supplements help prevent cancer or lower the chances of dying of cancer. In fact, some studies find that in people who smoke or used to smoke, high doses of beta-carotene supplements can increase the risk of lung cancer and death.

Age-related macular degeneration

Age-related macular degeneration (AMD) is the loss of central vision as people age. It's the most common cause of vision loss in older people. Studies show that a supplement containing vitamins C and E, zinc, and copper with or without beta-carotene helps slow down the rate of vision loss in people with AMD who are at high risk of developing advanced AMD. The same supplement, containing lutein and zeaxanthin instead of beta-carotene, reduces the risk of progression to advanced AMD even more and eliminates the increased risk of lung cancer from high doses of beta-carotene.

Measles

In developing countries where vitamin A deficiency is common, children with measles are more likely to have severe symptoms and may die from the disease. In these children, taking vitamin A supplements can help prevent new cases of measles but probably doesn't lower their risk of dying of measles.

Can vitamin A be harmful?

Yes, high intakes of some forms of vitamin A can be harmful.

Getting too much preformed vitamin A (usually from supplements or certain medicines) can cause severe headache, blurred vision, nausea, dizziness, muscle aches, and problems with coordination. In severe cases, getting too much preformed vitamin A can even lead to coma and death.

If you take too much preformed vitamin A while pregnant, it can cause birth defects in your baby, including abnormal eyes, skull, lungs, and heart. If you are or might be pregnant or breastfeeding, you should not take high-dose supplements of preformed vitamin A.

High intakes of beta-carotene do not cause the same problems as preformed vitamin A. Consuming high amounts of beta-carotene can turn the skin yellow-orange, but this condition is harmless and goes away when you eat less of it. However, several studies have shown that smokers, former smokers, and people exposed to asbestos who take high-dose beta-carotene supplements have a higher risk of lung cancer and death.

The daily upper limits for *preformed* vitamin A include intakes from all sources—food, beverages, and supplements—and are listed below. These levels do not apply to people who are taking vitamin A for medical reasons under the care of a doctor. There are no upper limits for beta-carotene and other forms of provitamin A.

Ages	Upper Limit
Birth to 12 months	600 mcg
Children 1–3 years	600 mcg
Children 4–8 years	900 mcg
Children 9–13 years	1,700 mcg
Teens 14–18 years	2,800 mcg

Ages	Upper Limit
Adults 19 years and older	3,000 mcg

Does vitamin A interact with medications or other dietary supplements?

Yes, vitamin A supplements can interact or interfere with medicines you take. Here are several examples:

- Orlistat (Alli, Xenical), a weight-loss drug, can decrease the absorption of vitamin A. This can cause low blood levels of vitamin A in some people.
- Acitretin (Soriatane), used to treat psoriasis, and bexarotene (Targretin), used to treat the skin effects of T-cell lymphoma, are made from vitamin A. Taking these medicines together with a vitamin A supplement could cause dangerously high levels of vitamin A in the blood.

Tell your doctor, pharmacist, and other health care providers about any dietary supplements and medicines you take. They can tell you if those dietary supplements might interact or interfere with your prescription or over-the-counter medicines or if the medicines might interfere with how your body absorbs, uses, or breaks down nutrients.

Vitamin A and healthful eating

People should get most of their nutrients from food and beverages, according to the federal government's *Dietary Guidelines for Americans*. Foods contain vitamins, minerals, dietary fiber, and other components that benefit health. In some cases, fortified foods and dietary supplements are useful when it is not possible to meet needs for one or more nutrients (for example, during specific life stages such as pregnancy). For more information about building a healthy dietary pattern, see the [Dietary Guidelines for Americans](#).

Where can I find out more about vitamin A?

- For more information on vitamin A
 - Office of Dietary Supplements (ODS) [Health Professional Fact Sheet on Vitamin A](#)
 - [Vitamin A](#), MedlinePlus
- For more information on food sources of vitamin A
 - USDA's [FoodData Central](#)
 - Nutrient List for vitamin A (listed by [food](#) or by [vitamin A content](#)), USDA
 - Nutrient List for beta-carotene (listed by [food](#) or by [beta-carotene content](#)), USDA
- For more advice on choosing dietary supplements
 - ODS [Frequently Asked Questions: Where can I purchase dietary supplements?](#)
- For information about building a healthy dietary pattern
 - [Dietary Guidelines for Americans](#)

Glossary

absorption

In nutrition, the process of moving protein, carbohydrates, fats, and other nutrients from the digestive system into the bloodstream. Most absorption occurs in the small intestine.

age-related macular degeneration

AMD. An eye disease that results in a loss of central, "straight-ahead" vision. AMD is the leading cause of vision loss in older Americans.

anemia

A condition in which the number of red blood cells in the blood, or the amount of hemoglobin in them, is lower than normal, causing a condition in which red blood cells are not able to supply enough

oxygen to all the tissues in the body. Hemoglobin is the substance in red blood cells that carries oxygen to your body's cells.

beta-carotene

A carotenoid found in carrots, cantaloupe, apricots, sweet potatoes, pumpkin, winter squash, mangos, collard greens, spinach, kale, broccoli, and other orange, red, and dark green fruits and vegetables.

carotenoids

A group of pigments that naturally occur in some plants, algae, and bacteria. They are responsible for the yellow, orange, and red colors in plant foods, such as carrots, red peppers, pumpkin, beets, and sweet potatoes. Carotenoids are also found in spinach, collards, kale, mustard greens, and other dark green leafy vegetables. Common carotenoids include alpha-carotene, beta-carotene, beta-cryptoxanthin, lycopene, lutein, and zeaxanthin..

celiac disease

An autoimmune disorder in which eating gluten (a protein found in wheat, rye, barley, and possibly oats) causes the immune system to damage the small intestine, making it unable to absorb nutrients. It is a genetic disease that sometimes becomes active for the first time after surgery, pregnancy, childbirth, viral infection, or extreme stress. Also called sprue.

cell

The individual unit that makes up the tissues of the body. All living things are made up of one or more cells, which are the smallest units of living structure capable of independent existence.

cholesterol

A substance found throughout the body. It is made by the liver and is an important component of cells. Cholesterol is also used to make hormones, bile acid, and vitamin D. Foods that come from animals contain cholesterol, including eggs, dairy products, meat, poultry and fish. High blood levels of cholesterol increase a person's chance (risk) of developing atherosclerosis and heart disease.

copper

In nutrition, a mineral the body needs (along with iron) to make red blood cells. Copper also helps keep the immune system, blood vessels, nerves, and bones healthy. Copper is found in some foods, including oysters and other shellfish, whole grains, beans, nuts, potatoes, organ meats, dark leafy greens, and dried fruits.

Crohn's disease

A long-lasting (chronic) disease that causes severe irritation in the gastrointestinal tract. It usually affects the lower small intestine (called the ileum) or the colon, but it can affect any part of the digestive tract from the mouth to the anus. It is painful, causing severe watery or bloody diarrhea, and may lead to life-threatening complications. Crohn's disease is a form of inflammatory bowel disease.

cystic fibrosis

A common inherited disease that causes the body to make thick, sticky mucus that builds up in the lungs and blocks the airways, leading to repeated serious lung infections. Mucus also blocks the pancreas, which stops digestive enzymes from reaching the intestines. Cystic fibrosis also causes very salty sweat, which can lead to dehydration, increased heart rate, tiredness, low blood pressure, and heat stroke.

dairy food

Milk and products made with milk, such as buttermilk, yogurt, cheese, cottage cheese, and ice cream.

deficiency

An amount that is not enough; a shortage.

diarrhea

Loose, watery stools.

dietary fiber

A substance in plants that you cannot digest. It adds bulk to your diet to make you feel full, helps prevent constipation, and may help lower the risk of heart disease and diabetes. Good sources of dietary fiber include whole grains (such as brown rice, oats, quinoa, bulgur, and popcorn), legumes (such as black beans, garbanzo beans, split peas, and lentils), nuts, seeds, fruit, and vegetables.

Dietary Guidelines for Americans

Advice from the federal government to promote health and reduce the chance (risk) of long-lasting (chronic) diseases through nutrition and physical activity. The Guidelines are updated and published every 5 years by the US Department of Health and Human Services and the US Department of Agriculture.

dietary supplement

A product that is intended to supplement the diet. A dietary supplement contains one or more dietary ingredients (including vitamins, minerals, herbs or other botanicals, amino acids, and other substances) or their components; is intended to be taken by mouth as a pill, capsule, tablet, or liquid; and is identified on the front label of the product as being a dietary supplement.

dose

The amount of medicine or other substance taken at one time or over a specific period of time.

drug

Any substance (other than food) that is used to prevent, diagnose, treat, or relieve symptoms of a disease or abnormal condition. Also, a substance that alters mood or body function or that can be habit-forming or addictive, especially a narcotic.

fat soluble

Able to be dissolved in fat.

fortified

When nutrients (such as vitamins and minerals) are added to a food product. For example, when calcium is added to orange juice, the orange juice is said to be "fortified with calcium". Similarly, many breakfast cereals are "fortified" with several vitamins and minerals.

immune system

A group of organs and cells that defends the body against infection, disease, and altered (mutated) cells. It includes the thymus, spleen, lymphatic system (lymph nodes and lymph vessels), bone marrow, tonsils, and white blood cells.

infant

A child younger than 12 months old.

infection

The invasion and spread of germs in the body. The germs may be bacteria, viruses, yeast, or fungi.

interaction

A change in the way a dietary supplement acts in the body when taken with certain other supplements, medicines, or foods, or when taken with certain medical conditions. Interactions may cause the dietary supplement to be more or less effective, or cause effects on the body that are not expected.

liver

A large organ located in the right upper abdomen. It stores nutrients that come from food, makes chemicals needed by the body, and breaks down some medicines and harmful substances so they can be removed from the body.

lot

A batch, or a specific identified portion of a batch, having uniform character and quality within specified limits; or, an amount produced in a unit of time or quantity.

lung

An organ in the chest that supplies oxygen to the body and removes carbon dioxide. There are two lungs in the body.

lymphoma

Cancer that develops in cells of your immune system, called lymphocytes. Lymphocytes are a type of white blood cell.

measles

A group of diseases of the respiratory tract caused by a virus. Measles is highly contagious and spreads easily from person to person through coughing or sneezing. Symptoms include fever, cough, red and irritated eyes, and a spreading rash. Serious complications include pneumonia, inflammation of the brain, and death. One form called German measles may cause birth defects in a fetus if a woman is infected early in her pregnancy.

microgram

μg or mcg. A unit of weight in the metric system equal to one millionth of a gram. (A gram is approximately one-thirtieth of an ounce.)

mineral

In nutrition, an inorganic substance found in the earth that is required to maintain health.

nausea

The uneasy feeling of having an urge to throw up (vomit).

nutrient

A chemical compound in food that is used by the body to function and maintain health. Examples of nutrients include proteins, fats, carbohydrates, vitamins, and minerals.

Office of Dietary Supplements

ODS, Office of Disease Prevention, Office of Director, National Institutes of Health, Department of Health and Human Services. ODS strengthens knowledge and understanding of dietary supplements by evaluating scientific information, stimulating and supporting research, disseminating research results, and educating the public to foster an enhanced quality of life and health for the US population.

pharmacist

A person licensed to make and dispense (give out) prescription drugs and who has been taught how they work, how to use them, and their side effects.

pneumonia

Inflammation of one or both lungs. Bacteria, viruses, fungi, parasites, other germs, and injury can cause the lungs to become inflamed and fill with fluid. Symptoms can appear suddenly, range from mild to severe, and may include fever, chills, chest pain, cough, shortness of breath, and difficult breathing.

Anyone can develop pneumonia, but it is especially dangerous in babies, older people, and people with weakened immune systems, lung disease, heart disease, or diabetes.

premature infant

A baby born before 37 weeks of pregnancy. Normally pregnancy lasts 42 weeks. Also called preterm infant and preemie.

prescription

A written order from a health care provider for medicine, therapy, or tests.

prevent

To stop from happening.

provitamin

A substance found in some foods that the body can use to make a vitamin. An example of a provitamin is beta-carotene, which the body uses to make vitamin A. Also called a vitamin precursor.

psoriasis

A chronic inflammatory disease in which the skin becomes swollen, red, and itchy, with patches of silvery-white scales.

red blood cell

A cell that carries oxygen to and removes carbon dioxide from all parts of the body.

respiratory

Having to do with breathing. The organs that are involved with breathing include the mouth, nose, throat (pharynx), voicebox (larynx), windpipe (trachea), air passages between the windpipe and lungs (bronchial tubes), and lungs.

retinol

A form of vitamin A found only in foods that come from animals. The body can use retinol to make retinal and retinoic acid (other forms of vitamin A). Retinol is found in some foods, including beef liver, whole eggs, whole milk, margarine, and some fortified food products such as breakfast cereals. Retinol is a retinoid. Also called preformed vitamin A.

Retinol Activity Equivalent

RAE. A measure of the content and activity of vitamin A in foods.

retinyl palmitate

The main form in which retinol (the form of vitamin A in foods that come from animals) is stored in the body.

risk

The chance or probability that a harmful event will occur. In health, for example, the chance that someone will develop a disease or condition.

sign

An indication of disease that can be seen and/or measured. Examples include high fever, high blood pressure, infection, and coughing up blood.

supplement

A nutrient that may be added to the diet to increase the intake of that nutrient. Sometimes used to mean dietary supplement.

treat

To care for a patient with a disease by using medicine, surgery, or other approaches.

ulcerative colitis

Chronic inflammation of the colon that causes ulcers to form in its lining. This condition is marked by abdominal pain, cramps, and loose discharges of pus, blood, and mucus from the bowel.

upper limit

UL. The largest daily intake of a nutrient considered safe for most people. Taking more than the UL is not recommended and may be harmful. The UL for each nutrient is set by the Food and Nutrition Board at the National Academies of Sciences, Engineering, and Medicine. For example, the UL for vitamin A is 3,000 micrograms/day. Women who consume more than this amount every day shortly before or during pregnancy have an increased chance (risk) of having a baby with a birth defect. Also called the tolerable upper intake level.

US Department of Agriculture

USDA promotes America's health through food and nutrition, and advances the science of nutrition by monitoring food and nutrient consumption and updating nutrient requirements and food composition data. USDA is responsible for food safety, improving nutrition and health by providing food assistance and nutrition education, expanding markets for agricultural products, managing and protecting US public and private lands, and providing financial programs to improve the economy and quality of rural American life.

vitamin

A nutrient that the body needs in small amounts to function and maintain health. Examples are vitamins A, C, and E.

vitamin A

A general term for a group of compounds that includes provitamin A carotenoids (found in foods that come from plants) and retinol (preformed vitamin A found in foods that come from animals). The body can use retinol to make retinal and retinoic acid (other forms of vitamin A). Vitamin A plays an important role in vision, bone growth, reproduction, immunity, cell development, and skin health. Vitamin A is found in some foods, including eggs, liver, fortified milk, cheese, leafy green vegetables (such as spinach, kale, turnip greens, collards, and romaine lettuce), broccoli, dark orange fruits and vegetables (such as apricots, carrots, pumpkin, sweet potatoes, papaya, mango, and cantaloupe), and red bell pepper.

zinc

A mineral found in most cells of the body. It helps enzymes work properly, helps maintain a healthy immune system, helps maintain the senses of taste and smell, and is needed for wound healing, making DNA, and normal growth and development during pregnancy, childhood, and adolescence. Zinc is found in some foods, including oysters, red meat, poultry, beans, nuts, certain seafood, whole grains, fortified breakfast cereals, and dairy products.

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